

ACHILA THILAKARATHNA

Mechatronics Engineer

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PROFILE SUMMARY

First Class BSc (Hons) graduate in Science and Technology, specializing in Mechatronics, with hands-on industrial experience in automation, machinery maintenance, PLC programming, pneumatic systems, electromechanical troubleshooting and manufacturing processes. Experienced in troubleshooting automated machinery, motion-control systems, electrical/electronic systems and industrial equipment within a production environment. Practical exposure to AGVs, RFID systems, stepper motors, pneumatic systems, solar-powered equipment, CAD/CAM and machine fabrication. Strong problem-solving skills with the ability to work in fast-paced manufacturing environments.

EDUCATION

BSc. Hons in Science and Technology - Specialised in Mechatronics

Uva Wellassa University of Sri Lanka
2021 - 2025

Graduated with highest honors, recognizing academic excellence.

GPA: 3.76/4.00

EXPERIENCE

Autonomation Engineer

March 2026 - October 2026
MAS Intimates Linear Clothing, Unichela (Pvt) Ltd.

- Maintained and troubleshot automated manufacturing machinery pneumatic, electromechanical, electrical, and sensor/actuator systems.
- Developed and modified PLC ladder logic for AGV, machine-control, RFID positioning, and stepper-motor motion control.
- Designed a new lap-laying machine for heated elastic to eliminate tangling and save production time; also maintained golf cart solar systems and fiber-optic audio connections.
- Supported machine design/fabrication (Fusion 360, RDWorks) with lathe/milling exposure; contributed to 5S and component organization.

Hardware Engineer

ExcelTech OU
June 2025 - December 2025

- Designed multilayer PCB schematics/layouts (Altium, custom libraries, DBLib) and performed assembly, SMD/SMT/THT soldering, and board-level troubleshooting.
- Developed embedded firmware for STM32, TI, and ESP32/ESP32-C3 (STM32CubeIDE, CCS, TI UniFlash, Arduino IDE); implemented UART, I²C, SPI, and RS-485 interfaces.
- Performed hardware validation, functional testing, and QA support using oscilloscopes and logic analyzers to improve system reliability.
- Built a wearable IMU device (bare-metal STM32), reverse-engineered a refrigerator control board, and designed a thermal-imaging camera module.

SKILLS

- PCB Designing (Altium Designer & Fusion 360 Electronics)
- 3D Modeling and Designing
- CAD/CAM (SolidWorks / Solid Edge / Fusion 360)
- PLC Programming
- Pneumatic Systems
- ROS2
- STM32/ESP32 Embedded Systems
- Firmware Programming
- C/C++/Python Coding
- Fiber Optic Communication Systems
- SMD/SMT/THT Soldering
- Robotics
- Team Leadership
- Problem Solving
- Solar PV Systems & Maintenance
- Project Management
- Public Relations
- Teamwork
- Time Management

REFERENCES

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